

SUDESH MITHAPELLI

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ABOUT ME

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📍 Pune - 411044

ACADEMIC BACKGROUND

Master's in Modeling and Simulation

(Computational Fluid Dynamics)

Center for Modeling and Simulation

Pune University GPA - 4.08/6

Project - Numerical Modeling of a Fixed Bed Adsorber

Bachelors in Mechanical Engineering

Pune University 1st class

Project - Design and Manufacturing of work holding fixtures

PROFICIENT SKILLS

- CFD Tools - ANSYS, FLOW3D, FLOW3D CAST, FLOW3D WELD, FlowSight, Enight
- Meshing Tools - ANSA, GAMBIT
- CAD Tools - RHINO, CATIA, I-DEAS, SOLIDWORKS, AUTOCAD, AutoLISP.
- Programming - PYTHON, MATLAB, C, C++ 11/14/17/20, Data Structures and Algorithms, STL
- Operating Systems - WINDOWS, LINUX
- Customer Support platform - Salesforce
- Azure DevOps

LANGUAGES FAMILIAR

English, Hindi, Marathi, Telugu

KEY AREAS WORKED ON

C++ Development, CFD, CAE, Heat transfer, Fluid Mechanics, Fluid Dynamics, Water, Hydraulics, Casting, Turbulence modeling, PyQt based Flow Sight (Post-processor and visualisation tool) GUI development, Technical support, Embedded, Linux driver programming, TCP/IP, Socket programming, Multi-threading, Git, OOPS, OOAD.

INTERESTS

Long Distance Running, Ultra-marathon, Marathon

SUMMARY

A dynamic, diligent, performance driven and highly motivated Engineer with 10 years of experience which includes Python Qt based post-processor GUI development, Python Automation for FlowSight and FLOW3D features in CFD/CAE domain, C++ Development and Automation for Telecommunication applications. Experience in Embedded, Linux driver programming, UART based serial communication, TCP/IP based Socket programming, Multi-threading

WORK EXPERIENCE

C/C++ Software Development - Jan 2024 to present
Emerson, Pune

- Embedded Linux Drivers programming(UART).
- TCP/IP Socket programming.
- C/C++ development, Python scripting.
- Developing Simulators for DeltaV product related applications.

Technical Lead (Mar 2020 to Jan 2024)

Tech Mahindra Pvt. Ltd., Pune

- Telecommunication application
- Lead in various modules
- C++ development, Python scripting & automation

CFD Engineer - Nov 2014 to Nov 2019

Flow Science Software Pvt. Ltd. – Pune

- Python Qt based FlowSight (Post-processor and Visualisation tool) GUI development.
- FLOW3D Customization in C++ and FORTRAN
- Develop and deliver training for FLOW3D CFD/CAE products to the customers.
- Develop test problems and procedures to ensure continuous improvements to FLOW3D CFD/CAE products.
- CFD/CAE products Solver Testing.

CFD Engineer - Aug 2013 to Nov 2014

Aprameya Associates – Pune

- CAD Modeling
- CFD Analysis of Electrostatic Precipitator & Bag house Filter.
- Numerical Simulation of Torpedo Firing Mechanism.

PROJECTS WORKED ON

★ EMERSON

- Simulator(serial communication Linux based)
 - Designing and developing from scratch a serial communication UART protocol based Simulator that Simulates/mimics competitors IO cards.
 - This is developed in Linux using Linux driver programming.
 - It has two parts: Application and the Kernel side code that maintains respective databases and update each other by opening and closing character devices.
 - This leads to mimic behaviour of the old IO cards(which are costly and tedious to replace in automation industries) and thereby showcase customers that they can adopt DeltaV product without replacing the competitors IO cards.
- Simulator(Network communication Windows based)
 - Designing and developing from scratch a Network communication(TCP/IP) based Simulator that Simulates/mimics competitors IO cards.
 - This is developed in Windows and involves reading data from a Gateway application SQLite database, receiving messages from Client and processing and sending responses from the Server end via TCP/IP communication(Socket programming).
 - This leads to mimic behaviour of the old IO cards(which are costly and tedious to replace in automation industries) and thereby showcase customers that they can adopt DeltaV product without replacing the competitors IO cards.

★ TECH MAHINDRA PRIVATE LIMITED

- Write programs and applications for a business unit on a C/C++ program.
- Guide junior colleagues for developing code as per the requirement specified.
- Perform testing, debugging and fixing applications using the C/C++ platform.
- Design, develop, test, implement and code solutions on C/C++ programs on windows and Unix.
- Review, analyse and assess programs, designs and specifications for own programs on C/C++ applications.
- Create integrated features on payroll enterprise/Telecommunication domain software solutions on SQL, Oracle, and other C/C++ programs.
- Communicate designs and implementations to the team members and the documentation department.
- Complete appropriate test plans and documentation using the C/C++ program.
- Perform modifications and break fixes with C/C++ program.
- Quickly analyse and interpret code through performing system support analysis.
- Complete projects on time and within the budget limit Ability to work and resolve issues independently.
- SVN to Git migration activity.
- Shell scripting.
- Support Testing team and Application Support Team.
- TCP/IP based Socket programming for various applications associated with the Telecommunication domain.

★ FLOW SCIENCE SOFTWARE PRIVATE LIMITED

➤ **Python Qt based FlowSight(Post-processor and Visualization tool) GUI development in Windows & Linux.**

- Develop various features and make them available in FlowSight GUI, like Right click menus.
- Add Modify dialogs for existing features.
- Add new widgets and inculcate new features through it.
- Modify existing features.
- Coordinate with the Testing team to train them for Testing the post-processing features available in FlowSight.
- Automate and add processes like FFT calculations at the flick of a button in FlowSight GUI using Python numpy library.

➤ **Python Automation(Windows)**

- Automate generation of material data files using Python for FLOW3D Material database.
- Set up or write automated test scripts in Python and Squish for GUI Testing.
- Modify existing Python test scripts as per the changes in the GUI.

➤ **CFD/CAE Solver Testing(Windows & Linux)**

- Testing of the new models developed and about to be inculcated in the next cycle of FLOW-3D release.
- Collaborating with GUI and Solver developer teams.

➤ **Continually learn a wide range of physics and numerical methods applied to existing and newly developed models.**

- Learn, train and teach new Physics and Numerics based models.

➤ **Develop test problems and procedures to ensure continuous improvements to FLOW-3D CFD /CAE products.**

- Know problems faced by customers in the current FLOW3D version.
- Optimise and set up small test cases to help Developers understand the problem.
- Meanwhile, suggest alternatives to the customers to get their job done.
- Provide improvised solutions to the customers after thorough Testing.

➤ **Collaborate with the Marketing and Sales team to demonstrate and promote the products to the potential customers.**

- Set up test cases as a part of Proof of concept for demonstration purposes.
- Set up automated running of Simulations in batch mode for conferences.
- Develop test cases continuously on Verification and Validation projects to create a database for showcase purposes.

➤ **Develop and deliver training for FLOW-3D to the customers either in-house or on- site for a better and efficient usage of FLOW-3D CFD/CAE products.**

- Creating FlowSight recordings for uploading on the User site.
- Updating FLOW3D, FlowSight user manual as per the release work.

➤ **FLOW3D Customization (in C++)**

- Updating FLOW3D, FlowSight user manual as per the release work.
- Implement customizations for FLOW3D customers in C++ and FORTRAN and try inculcating the new customizations as a new functionality in the future release.

➤ **Provide Technical support and guidance to the customers with their complex and challenging 3D simulations in FLOW-3D CFD/CAE products.**

- Client onboarding
- Troubleshooting
- Efficient simulation setup
- Correct usage of the Physics models available in FLOW3D.
- Efficient post-processing
- Training via live video meetings, emails or on Telephone.

★ **APRAMEYA ASSOCIATES**

➤ **CFD Analysis of Electrostatic Precipitators**

- Receiving Inputs from client
- Reading 2D drawings in AutoCAD
- Creating 3D models in I-DEAS.
- Meshing in GAMBIT.
- Pre-process, Run Simulations and Post-process in ANSYS CFX.
- Implement design variations for various flow guiding vanes.
- Prepare presentations to communicate results from Analysis to the customers.

➤ **Numerical Simulation of Torpedo firing mechanism on Windows.**

- Develop an optimised numerical tool which will suggest Initial Pressure for firing Torpedo from submarine as per the depth of submarine operation.
- Programming in MATLAB.

➤ **Developing User defined functions in C++ for Simulations.**

❖ **M.TECH PROJECTS**

➤ **Develop a Numerical Model for a Fixed Bed Adsorber(Final year project) on Windows.**

- Developed Numerical model that mimics behaviour of a fixed bed adsorber used in water filter cartridges.
- Programming and Computing in MATLAB.
- Discretized Partial difference equations and the CFD techniques to solve the governing equation of the model.
- Solved equations using both Implicit and Explicit methods.

➤ **Implemented Game theory Application on Windows and Linux based environments.**

- Programming in C++.
- Object oriented programming.

➤ **Implement Numerical methods in C++ (Windows & Linux environment)**

❖ **B.E. PROJECT**

➤ **Design and Manufacturing of work holding fixtures.**

- Developed a new light weight fixture design.
- Used ALUMEC for reduction in weight.
- Provided optimised clamping.
- Reduced work load on the pneumatic systems.

Certifications done:

Data Structures and Algorithms

Source - Udemy

Programming in C++

Source - NPTEL(IIT MADRAS)

Modern C++ 11/14/17

Source - Udemy